CSC 220 Data Structures & Program Design - Syllabus

Bay State Community College
Credits: 4
Instructor: Course staff Office hours: by appointment

Course Description
Data structures and program design: implementation and analysis of fundamental structures and the algorithms that use them.

Learning Outcomes
- Implement lists, stacks, queues, trees and hash tables
- analyse running time using Big-O notation
- design object-oriented programs

Topics Covered
- Arrays, linked lists, stacks and queues
- Trees and binary search trees
- Hash tables and sets
- Sorting and searching algorithms
- Algorithm efficiency and Big-O analysis

Weekly Schedule
Week 1-3: Arrays, linked lists, stacks and queues
Week 4-6: Trees and binary search trees
Week 7-9: Hash tables and sets
Week 10-12: Sorting and searching algorithms
Week 13-15: Algorithm efficiency and Big-O analysis

Grading Policy
Assignments 30%, midterm exam 30%, final exam 40%. Prerequisites: as listed in the catalogue. Academic integrity policy applies.